The FIM is invariant by reparameterization of the sample space $\mathcal{X}$, and covariant by reparameterization of the parameter space Θ, see [25]. That is, let $\bar{p}(x;\eta) = p(\theta(\eta);x)$. Then we have:

$$\bar{I}(\eta) = \left[\frac{\partial \theta_i}{\partial \eta_j}\right]_{ij}^{\top} I(\theta(\eta)) \left[\frac{\partial \theta_i}{\partial \eta_j}\right]_{ij}. \tag{81}$$

Matrix $J_{ij} = \left[\frac{\partial \theta_i}{\partial \eta_j}\right]_{ij}$ is the Jacobian matrix.

Example 1. *For example, consider the family*

$$\mathcal{N} = \left\{ p(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x-\mu)^2}{2\sigma^2}) \ : \ (\mu,\sigma) \in \mathbb{R} \times \mathbb{R}_{++} \right\} \tag{82}$$

of univariate normal distributions. The 2D parameter vector is $\lambda = (\mu,\sigma)$ with μ denoting the mean and σ the standard deviation. Another common parameterization of the normal family is $\lambda' = (\mu,\sigma^2)$. The λ' parameterization extends naturally to d-variance normal distributions with $\lambda' = (\mu,\Sigma)$, where Σ denotes the covariance matrix (with $\Sigma = \sigma^2$ when $d = 1$). For multivariate normal distributions, the λ-parameterization can be interpreted as $\lambda = (\mu, L^{\top})$ where $L^{\top}$ is the upper triangular matrix in the Cholesky decomposition (when $d = 1$, $L^{\top} = \sigma$). We have the following Fisher information matrices in the λ-parameterization and λ'-parameterization:

$$I_{\lambda}(\lambda) = \begin{bmatrix} \frac{1}{\lambda_2^2} & 0 \\ 0 & \frac{2}{\lambda_2^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{bmatrix} \tag{83}$$

and

$$I_{\lambda'}(\lambda') = \begin{bmatrix} \frac{1}{\lambda_2} & 0 \\ 0 & \frac{1}{2\lambda_2^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix} \tag{84}$$

Since the FIM is covariant, we have the following the change of transformation:

$$I_{\lambda'}(\lambda') = J_{\lambda,\lambda'}^{\top} I_{\lambda}(\lambda(\lambda')) J_{\lambda,\lambda'}, \tag{85}$$

with

$$J_{\lambda',\lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 2\sigma \end{bmatrix} \tag{86}$$

Thus we check that

$$I_{\lambda}(\lambda) = \begin{bmatrix} 1 & 0 \\ 0 & 2\sigma \end{bmatrix} \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2\sigma \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{bmatrix} \tag{87}$$

Notice that the infinitesimal length elements are invariant: $\mathrm{d}s_{\lambda} = \mathrm{d}s_{\lambda'}$.

As a corollary, notice that we can recognize the Euclidean metric in any other coordinate system if the metric tensor g can be written $J_{\lambda,\lambda'}^{\top} J_{\lambda,\lambda'}$. For example, the Riemannian geometry induced by a dually flat space with a separable potential function is Euclidean [49].

In statistics, the FIM plays a role in the attainable precision of unbiased estimators. For any unbiased estimator, the Cramér-Rao lower bound [71] on the variance of the estimator is:

$$\mathrm{Var}_{\theta}[\hat{\theta}_n(X)] \succeq \frac{1}{n} {}_{\mathcal{P}}I^{-1}(\theta). \tag{88}$$

Figure 8 illustrates the Cramér-Rao lower bound (CRLB) for the univariate distributions: At regular grid locations (μ,σ) of the upper space of normal parameters, we repeat 200 runs (trials) of estimating the normal parameters $\widehat{(\mu,\sigma)}$ using the MLE on 100 iid samples $x_1,\ldots,x_n \sim N(\mu,\sigma)$. The sample mean and the sample covariance matrix are calculated for the number of trials and displayed as back ellipses. The Fisher

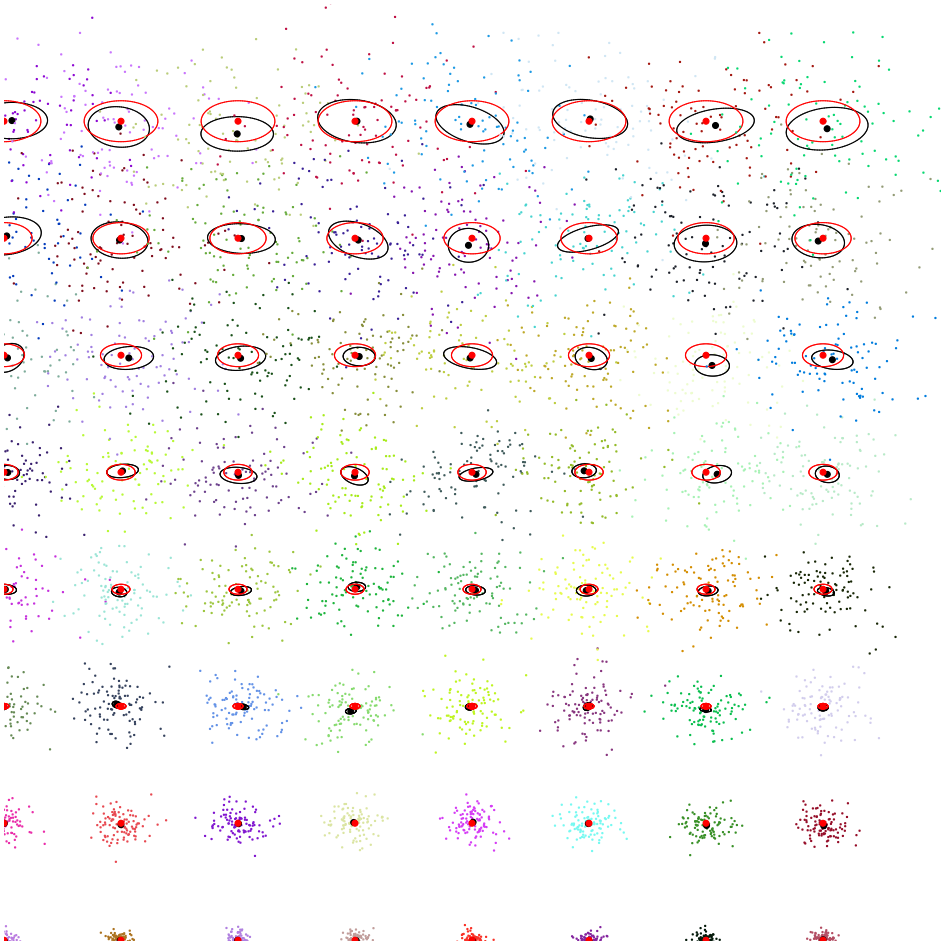

Figure 8: Visualizing the Cramér-Rao lower bound: The red ellipses display the Fisher information matrix of normal distributions at grid locations. The black ellipses are sample covariance matrices centered at the sample means calculated by repeating 200 runs of sampling 100 iid variates for the normal parameters of the grid.

information matrix is plotted as red ellipses at the grid locations: The red ellipses have semi-axis parallel to the coordinate system since the parameters μ and σ are orthogonal (diagonal FIM). This is not true anymore for the sample covariance matrix of the MLE estimator, and the centers of the sample covariance matrices deviate from the grid locations.

We report the expression of the FIM for two important generic parametric family of probability distributions: (1) an exponential family (with its prominent multivariate normal family), and (2) a mixture family.

Example 2 (FIM of an exponential family $\mathcal{E}$)**.** *An* exponential family *[84] $\mathcal{E}$ is defined for a sufficient statistic vector $t(x) = (t_1(x), \ldots, t_D(x))$, and an auxiliary carrier measure $k(x)$ by the following canonical density:*

$$\mathcal{E} = \left\{ p_\theta(x) = \exp\left(\sum_{i=1}^{D} t_i(x)\theta_i - F(\theta) + k(x) \right) \text{ such that } \theta \in \Theta \right\}, \tag{89}$$

where F is the strictly convex cumulant function (also called log-normalizer, and log partition function or free energy in statistical mechanics). Exponential families include the Gaussian family, the Gamma and Beta families, the probability simplex Δ, etc. The FIM of an exponential family is given by:

$${}_{\mathcal{E}}I(\theta) = \mathrm{Cov}_{X\sim p_\theta(x)}[t(x)] = \nabla^2 F(\theta) = (\nabla^2 F^*(\eta))^{-1} \succ 0. \tag{90}$$

Natural parameters beyond vector types can also be used in the canonical decomposition of the density of an exponential family: For example, we may use a matrix type for defining the zero-centered multivariate Gaussian family or the Wishart family, a complex numbers for defining the complex-valued Gaussian distribution family, etc. We then replace the term $\sum_{i=1}^{D} t_i(x)\theta_i$ in Eq. 89 by an inner product defined for the natural parameter type (e.g., dot product for vectors, matrix product trace for matrices, etc). Furthermore, natural parameters can be of compound types: For example, the multivariate Gaussian distribution can be written using $\theta = (\theta_v, \theta_M)$ where θ_v is a vector part and θ_M a matrix part, see [84].

Let $\Sigma = [\sigma_{ij}]$ denote the covariance matrix *and $\Sigma^{-1} = [\sigma^{ij}]$ the* precision matrix *of a multivariate normal distribution. The Fisher information matrix of the multivariate Gaussian [114, 121] $N(\mu, \Sigma)$ is given by*

$$I(\mu,\Sigma) = \begin{array}{c} \begin{array}{cc} \mu & \Sigma = [\sigma_{ij}] \end{array} \\ \begin{bmatrix} \sigma^{ij} & 0 \\ 0 & \sigma^{il}\sigma^{jk} + \sigma^{ik}\sigma^{jl} \end{bmatrix} \end{array} \begin{array}{c} \mu \\ \Sigma = [\sigma_{kl}] \end{array} \tag{91}$$

Notice that the lower right block matrix is a 4D tensor of dimension $d \times d \times d \times d$. The zero subblock matrices in the FIM indicate that the parameters μ and Σ are orthogonal to each other. In particular, when $d = 1$, since $\sigma^{11} = \frac{1}{\sigma^2}$, we recover the Fisher information matrix of the univariate Gaussian:

$$I(\mu,\Sigma) = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix} \tag{92}$$

We refer to [63] for the FIM of a Gaussian distribution using other canonical parameterizations (natural/expectation parameters of exponential family).

Example 3 (FIM of a mixture family $\mathcal{M}$)**.** *A mixture family is defined for $D+1$ functions $F_1, \ldots, F_D$ and C as:*

$$\mathcal{M} = \left\{ p_\theta(x) = \sum_{i=1}^{D} \theta_i F_i(x) + C(x) \text{ such that } \theta \in \Theta \right\}, \tag{93}$$

where the functions $\{F_i(x)\}_i$ are linearly independent on the common support $\mathcal{X}$ and satisfying $\int F_i(x)\mathrm{d}\mu(x) = 0$. Function C is such that $\int C(x)\mathrm{d}\mu(x) = 1$. Mixture families include statistical mixtures with prescribed component distributions and the probability simplex Δ. The FIM of a mixture family is given by:

$${}_{\mathcal{M}}I(\theta) = E_{X\sim p_\theta(x)}\left[\frac{F_i(x)F_j(x)}{(p_\theta(x))^2} \right] = \int_{\mathcal{X}} \frac{F_i(x)F_j(x)}{p_\theta(x)} \mathrm{d}\mu(x) \succ 0. \tag{94}$$